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An adaptive heuristic algorithm with a collaborative search framework for multi-UAV inspection planning

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HIGHLIGHTS

- A region gridding disperse approach with minimum redundancy time which enhances the flexibility of UAVs path planning.
- An adaptive hybrid initialization method which can effectively improve the algorithm's efficiency and stability.
- A swarm intelligence optimization algorithm with a novel collaborative search framework to solve the multi-UAVs planning problems effectively.

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ABSTRACT

Multi-UAV inspection path planning has become an important research topic for completing inspection tasks before the data acquisition deadline. In this study, we propose an adaptive heuristic algorithm with a collaborative search framework named Sa-VCO to solve the multi-UAV inspection path planning problem. Our study includes three main novelties. First, we design a region-gridding disperse approach that transforms the primitive target regions into a set of standard target subregions, allowing the target regions with greater costs to be inspected by multiple UAVs. Second, we propose an adaptive initial solution generation strategy using the information of graph structure constructed by all targets to reduce redundant computing. Third, we established a collaborative search framework to enhance search efficiency and increase population diversity. A large number of multiple-perspective comparative experiments are provided to test Sa-VCO's performance, and the comparison results demonstrate that Sa-VCO achieves better results than other advanced algorithms, especially on large-scale data sets.

1. Introduction

The inspection problem aims to complete a series of specified data acquisition tasks for a set of determined target regions, which widely exists in various application fields such as disaster rescue [1], intelligent agriculture [2], environment protection [3], wireless sensor networks [4], etc. Considering the data to be collected in most application scenarios has timeliness, the inspection tasks are expected to be completed in the shortest possible time.

In recent years, benefiting from the rapid development of computer science, electronics, and machinery technologies, unmanned aerial vehicles (UAVs) have been introduced as efficient equipment for inspection tasks. UAVs have unique air superiority to avoid interference by complex terrain conditions and flexible ability to carry different detection sensors. Still, they are also limited by endurance, capacity, and vulnerability to weather [5]. Researchers in relevant fields expect to make full use of their advantages and weaken the impact of their disadvantages as much as possible. As a result, the multi-UAV path planning problem has become a hot research topic.

The multi-UAV inspection path planning can be mainly divided into

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1568-4946/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

two stages: the global coverage paths planning stage and the inspection trajectories planning stage [6]. In the first stage, the targets are evenly assigned to different drones, and a rough flight path is planned for each drone without considering the interference of terrain, obstacle avoidance, etc. Then, in the second stage, the three-dimensional UAV flight trajectory with security guarantee will be further planned for each path segment (defined by each two of adjacent targets which have a connection relationship in the rough planning paths) according to the actual terrain.

Related research has indicated that the quality of the first-stage solution will significantly affect the inspection efficiency and the computational complexity of this stage is NP-hard [7]. This means that when the number of targets and drones increases, solving this problem will consume huge computing resources. The existing methods can hardly obtain satisfactory results, and we note that the research on multi-UAV path planning is mostly focused on the three-dimensional UAV flight trajectories planning stage, but less on the global path planning stage. In particular, there is a lack of methods for large-scale multi-UAV inspection global path planning problems.

This paper is devoted to solving the large-scale multi-UAV inspection global path planning problem under acceptable time costs and computational costs. First, we note that in the practical application scenario of the multi-UAV inspection problem, the area of the target areas is often uneven. Taking the application of emergency search and rescue as an example, as the trapped people will not be evenly distributed throughout the affected area, rescuers generally do not need to inspect the entire disaster area but only multiple sub-regions where people may gather. Due to the difference in the numbers of gathered people and the reasons for the people gathered, the areas of these areas are often different as well. In extreme cases, the area of a larger area may be dozens of times larger than that of a smaller area. This means it is not appropriate to use the target area as the minimum inspection unit (any area can only be assigned to a single UAV for inspection). This study proposed a region-gridding disperse approach with minimum redundancy area to balance the volume of task assignments better and enhance inspection flexibility. This method discretizes the original target areas into a set of standard target center points which allows the target areas with much higher inspection costs to be inspected by multiple drones. As a result, it effectively enhances the inspection efficiency and execution ability of UAVs under the same resource conditions.

Second, we find that our proposed region-gridding dispersal method may increase the number of targets (ten times higher than the initial target areas in extreme cases) and will exacerbate the difficulty of solving. To deal with this, this study proposed an efficient global path planning algorithm Self-adaptive Virus Control Optimizer (Sa-VCO) based on swarm intelligence. The main innovation of Sa-VCO is mainly reflected in two aspects. First, we designed an "efficiency-quality" balanced self-adaptive initial solution generation strategy to improve search efficiency and reduce blind exploration in the early stage. Second, we focus on the specific characteristics of the multi-UAV path planning problem. That is, the complex search space caused by the high coupling relationship between the target assignment within all sub-paths and the inspection order determined for each sub-path. We established a heuristic swarm collaborative search framework with a restricted multi-subpopulation information interaction mechanism and designed three novel search strategies under this framework. In addition, we defined the mathematical model for the multi-UAV cooperative inspection global path planning problem. Multiple sets of simulation tests are organized to verify the effectiveness of our proposed method.

In general, the contributions of this paper can be summarized as follows:

1) We defined the mathematical model for the multi-UAV cooperative inspection global path planning problem.
2) We propose a region gridding disperse approach with minimum redundancy area which can effectively enhance the flexibility of both UAV task allocation and UAV path planning.
3) We propose a swarm intelligence optimization algorithm Sa-VCO with an adaptive initial population generation method and a novel collaborative search framework based on subpopulations to effectively solve the multi-UAV inspection planning problems.
4) We set up multiple sets of simulation experiments to test the effectiveness of the proposed method and compare it with a variety of advanced algorithms. The results show that Sa-VCO has significant advantages in multi-UAV inspection global path planning, especially in large-scale problems.

The remainder of this study is structured as follows. 2 summarizes the recent related literature. 3 presents the problem's mathematical model and introduces the regional discretization method designed in this study. 4 describes the proposed algorithm's operational mechanism in detail. 5 provides experimental results comparing our approach to other studies using various test sets of different dimensions and distributions. Finally, 6 concludes the study and suggests future research directions.

2. Literature review

In this section, we analyze the related literature in detail to better understand the development trend of the multi-UAV inspection path planning. As mentioned above, this problem can be divided into two stages with different mathematical models of global coverage paths planning and inspection trajectories planning.

Most of the studies were carried out for the second stage. For example, Ge et. al. proposed a GWFOA algorithm for multi-UAV oilfield inspection path planning, which effectively overcame the problem of obstacle avoidance in the basic GWO algorithm and improved the path planning performance [8]. Wang et. al. introduced a modified Mayfly Algorithm for UAV three-dimensional flight trajectory planning in obstacle environments, which employs an adaptive Cauchy mutation strategy and an enhanced crossover operator to discover the path with the lowest overall cost [9]. Xiang et. al. considered the influence of variable size on computational efficiency and proposed EPSO-GA. This algorithm innovatively employed the Q-learning method and random strategy to obtain better initial paths [10]. Han et. al. focus on UAV indoor path planning and propose the grid-optimized A* path planning (GO-APP) algorithm and the grid-optimized local backtracking path planning (GO-LBPP) algorithm to address this problem. The proposed algorithms are verified to have shorter computation time and can obtain better flight trajectories by testing on multiple experiments [11]. Ali et. al. focus on the UAV three-dimensional flight trajectory planning problem with longer flight paths. Their study notes that as the number of planning points increases, the scale of decision variables rises and sharply increases the solving difficulty and required computing resources. Therefore, a decision model based on feature selection is designed to support the A* planner, which effectively improves planning efficiency [12]. Based on the relevant literature, we find that the research on UAV trajectory planning has comprehensively considered the collision threat (including collisions with obstacles or other drones) [13,14], the battery capacity [15,16], the system fault tolerance [17], the data capacity [18], the weather effect [7], etc.

These studies have greatly improved the application of UAVs in inspection missions. However, compared with the second stage, the quality of the first stage solution has a more decisive impact on the UAV inspection capability. There are also some existing studies devoted to obtaining a better solution for the global coverage path planning stage. Zhang et. al. aimed to maximize the total reconnaissance intelligence and proposed an improved harmony search algorithm [19]. Cheng et. al. established a task assignment model with multiple bases and rechargeable endurance for the multi-UAV road patrol problem and introduced

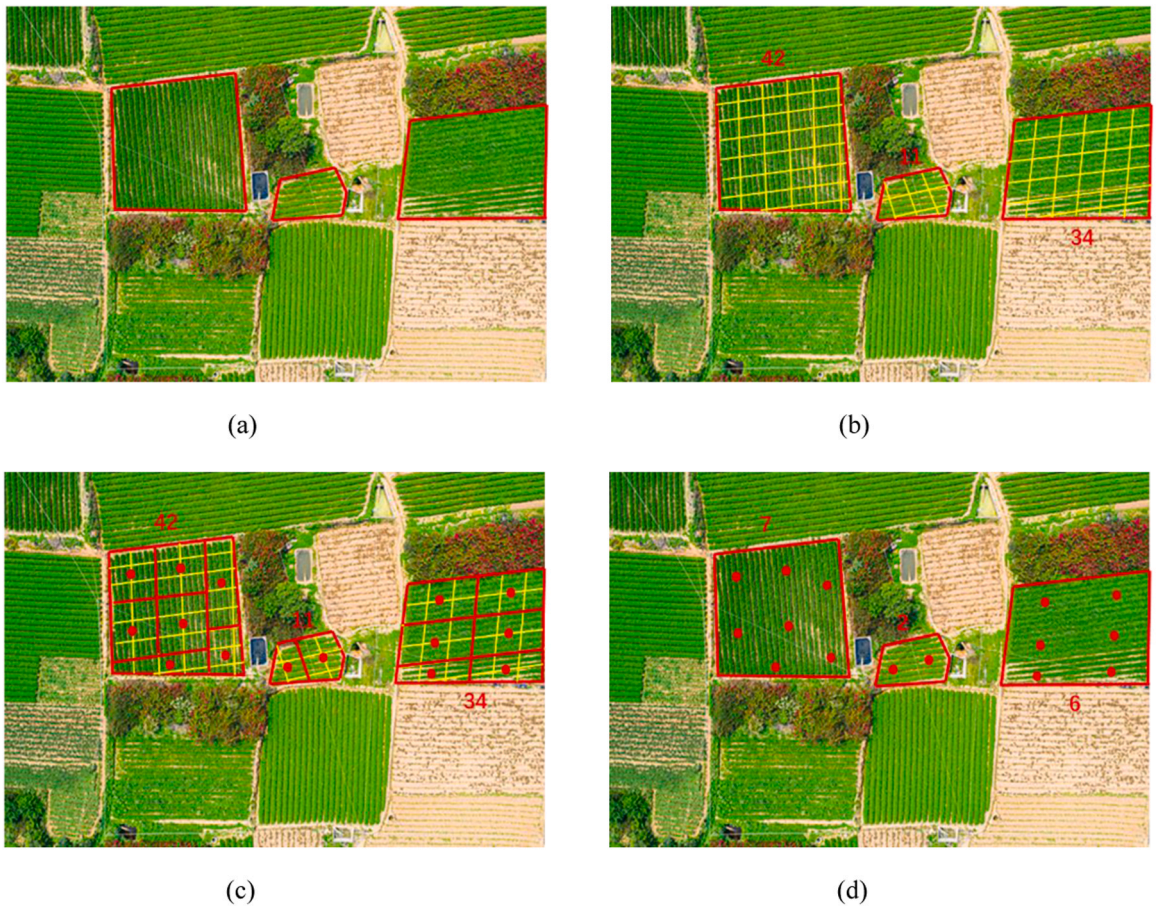


Fig. 1. The region discretization process.

an immune clonal selection algorithm for model solving. The effectiveness of the proposed method is verified by testing on multiple test sets including 20, 50, and 100 targets [20]. Cui et. al. proposed an opposition-based learning parameter-adjusting harmony search algorithm for multi-UAV task assignment in actual combat [21]. Huo et. al. improved the original Vehicle Routing Problem model (VRP) by adding virtual nodes according to the number of UAVs and proposed a Swap-and-Judge Simulated Annealing (SJSA) algorithm [22]. Liu et. al. aim to minimize the maximum flight distance of the UAV in autonomous inspection and proposed a two−step iterative heuristic algorithm. The simulation experiments of this study are taken on multiple Traveling Salesman Problem (mTSP) test sets, and the proposed algorithm overperformed the Genetic Algorithm (GA) [23]. Yan et. al. focus on the uneven task assignment in multi-UAV three-dimensional path planning and propose an improved fuzzy C-clustering method for UAV task allocation and a particle swarm hybrid ant colony (PSOHAC) algorithm for each UAV's path planning in its task area [24]. Li et. al. proposed an improved Bidirectional Ant Colony and discrete Honey Badger Algorithm (BACOHBA) to improve the quality of UAV global path planning solutions and thus effectively improve the UAV performance in transmission line inspection [7]. From the above literature, we can find that although there are already studies on UAV global coverage path planning, they mostly focus on small-scale and middle-scale problems. That is, the number of test targets is usually less than one thousand or even one hundred. When the number of targets continues to rise, these algorithms will consume a lot of computing time and can hardly find an optimal solution due to the limitation of the 'dimension disaster'.

3. Model description and region gridding disperse approach design

3.1. Problem definition

For the multi-UAV inspection application, we first consider an execution area (all targets are set in this area and the drones are not allowed to fly out), a starting depot, a set of target regions with different areas and shapes, and a set of drones with the same technical specifications and carrying the same sensors. This study does not consider multiple types of drones or drones starting from different initial depots, and information on all locations in all target regions should be collected to meet the inspection requirements. During the inspection process, all UAVs should take off from the starting depot, visit all target areas, and back to the depot. We take the time when the last drone returns to the depot as the inspection completed time.

This study assumes that all drones carry a camera as the data acquisition equipment, and the camera's height and angle relative to the ground target are considered fixed during inspection. We agree that the elevation changes of the ground should not be ignored to plan a feasible flight path. However, when this change does not exceed a certain limit, it will not bring additional energy consumption. Therefore, in the global coverage paths planning stage, we assume that all drones fly at a fixed speed and height, and we will not plan these two variables during the planning process (The actual flight speed and height will be planned in the inspection trajectories planning stage).

For the camera carried by the drones, we assume they can obtain information in a moving state and their shooting range is rectangular. As long as the ground target is within the drone shooting range at a time, the information at this target is considered to be collected. In summary,

for each target region, the drone assigned to inspect it should ensure that each position in this region has been within the range of its sensor at a certain moment, and the corresponding shooting areas of the UAV at different moments are not overlapped as much as possible.

Related studies mostly plan the global inspection path by first calculating the task volume of all target regions (UAV flight energy consumption) and then assigning these regions to different UAVs according to their task volume and location [25,26]. This method is usually effective and does not produce excessive computational requirements when the task size is not large. However, when the inspection mission contains a target region that has a large area (high task volume) and is far from the starting depot, the completion time of inspection will be decided by the UAV that is assigned to inspect this region. In extreme cases, a single UAV cannot independently complete the inspection task even if it is only designated to inspect this region.

To overcome this limitation, this study further standardizes the regions with different areas, so that the target regions with large task volumes can be assigned to multiple UAVs for collaborative inspection, thereby improving the inspection efficiency.

3.2. Region gridding disperse approach

In this study, we employed a grid-based approach to divide the inspection region into smaller standard subregions, each of which will be considered a separate target. The specific implementation steps of this method include:

Step 1: Set the initial grid size as the camera's shooting range at a unit time and calculate the number of discrete grids for all target regions.

Step 2: Find the maximum common factor for the grids contained in all target regions and update the standard subregions area as the sum of the area of the maximum common factor initial grid.

Step 3: Taking into account possible extreme situations, such as the existence of a very small region in the target regions, the above method may lead to hundreds of times increase in the number of standard subregions compared to the original target regions or even more. Therefore, to reduce the computational scale of the planning problem, the UAV is allowed to carry out a small amount of redundant inspection. The grid size setting will be taken as a multi-objective optimization problem, that is, to minimize the number of standard sub-regions and the area of redundant inspection regions simultaneously. The standard subregion size will finally be determined through iterative optimizations.

Step 4: We choose the standard subregions' center as the target points for the global coverage path planning stage, and the drone's actual flight trajectories for inspecting in the subregions are not considered during this stage. We only decide the task allocation and inspection sequence of the standard subregions.

The grid selection process is illustrated in Fig. 1.

Here, the regions marked by the red box in (a) are the target regions. The areas divided by yellow lines in (b) are the initial grids. We can find that there are 42, 11, and 34 independent grids in the three target regions. The maximum common factor of the initial grid number between these regions is 1. Without considering the consumption of redundant inspections, the number of inspection targets to be planned will rise from 3 to 87. By further optimizing the grid size, we regard each six initial grids as a standard subregion. At this time, the number of targets is reduced to 15 and only brought the additional energy consumption of inspecting 3 grids.

3.3. Model formulation

In summary, after converting the inspection regions into standard subregions, the multiple region coverage global path planning problem can be changed into the inspection planning for the set of discrete center points. As mentioned above, the energy consumption of inspecting all sub-regions is considered to be consistent. Therefore, the planning problem can be described by the multiple Traveling Salesman Problem (mTSP) model.

The mTSP is an extension of the famous Traveling Salesman Problem (TSP) and is recognized as a challenging Combinatorial Optimization Problem (COP) with NP-hard complexity. MTSP defines a set of salesmen and a set of cities to be visited, with the constraint that each city must be visited exactly once by a specific salesman and each salesman should visit at least one city. Minimizing the total path of all traveling salesman and minimizing the longest path among all traveling salesman are the two main optimization objectives of the problem. In this study, we define each drone as a salesman and each standard subregion as a city. We aim to minimize the path of a drone with the longest flight path, that is, to minimize the completion time of inspection tasks. On this basis, the total flight path of all UAVs is minimized to further reduce the multi-UAV system's energy consumption. Its mathematical representation is as follows:

$$\mathrm{Cos}\, t_{total} = \sum_{k=1}^{m}\sum_{i=1}^{n}\sum_{j=1;i\neq j}^{n}(d_{ij}\times x_{ij}^{k}) \tag{1}$$

$$\mathrm{Cos}\, t_{\max} = \max\Big(\sum_{i=1}^{n}\sum_{j=1;i\neq j}^{n}(d_{ij}\times x_{ij}^{k})\Big)\ \ \forall k=1,2,\cdots,m \tag{2}$$

Subject to:

$$\sum\nolimits_{i=1;i\neq j}^{n}\sum_{k=1}^{m}x_{ij}^{k}=1\ \ \forall j=1,2,\cdots,n \tag{3}$$

$$\sum\nolimits_{i=1;i\neq p}^{n}x_{ip}^{k}-\sum\nolimits_{j=1;j\neq p}^{n}x_{pj}^{k}=0\ \ \forall p=1,\cdots,n,\ \ k=1,2,\cdots,m \tag{4}$$

$$\sum_{i=1}^{n}\sum_{j=1;i\neq j}^{n}x_{ij}^{k}\geq 1\ \forall k=1,2,\cdots,m \tag{5}$$

$$u_i-u_j+Lx_{ij}\leq L-1\ \forall i,j=1,2,\cdots,n:i\neq j \tag{6}$$

$$1\leq u_i\leq L\ \forall i=1,2,\cdots,n \tag{7}$$

$$x_{ij}^{k}\in\{0,1\}\ \forall i,j=1,\cdots,n:i\neq j,k=1,2,\cdots,m \tag{8}$$

Here, (1) represents the total path cost of all salesmen and (2) represents the max path cost of all salesmen. (3)-(8) are the model constraints, (3) ensures that each city is visited exactly once by a UAV. (4) ensures that the k_{th} salesman who visits the p_{th} city must also leave the p_{th} city. (5) ensures that each salesman should visit at least one city. (6) and (7) limit the number of cities visited by any salesman, ensuring it does not exceed the maximum allowed by a sub-travel. (8) is the decision variable presenting whether the k_{th} salesman will leave the i_{th} city and then visit the j_{th} city. In these equations, m represents the number of salesmen (UAVs), n represents the number of cities (target points), d_{ij} represents the distance from the i_{th} city to the j_{th} city, and x_{ij}^{k} represents if the k_{th} salesman will travel from the i_{th} city to the j_{th} city. In MTSP solving, there are generally two optimization objectives: (1) minisum objective and (2) minimax objective. For the minisum optimization, the goal of problem-solving is to minimize the total path cost of all salesmen (mostly explained as to minimize the fuel used by all vehicles in applications, Eq. 1), and for the minimax optimization, the goal is to minimize the max path cost in all salesmen (mostly explained as to minimize the inspection time used by the vehicles with the longest time in applications, Eq. 2).

In this study, we take the minimax objective as the main optimization objective and the minisum objective as the secondary optimization objective to optimize the problem at the same time.

4. Sa-VCO algorithm

Based on the previous analysis, the key to the problem can be found

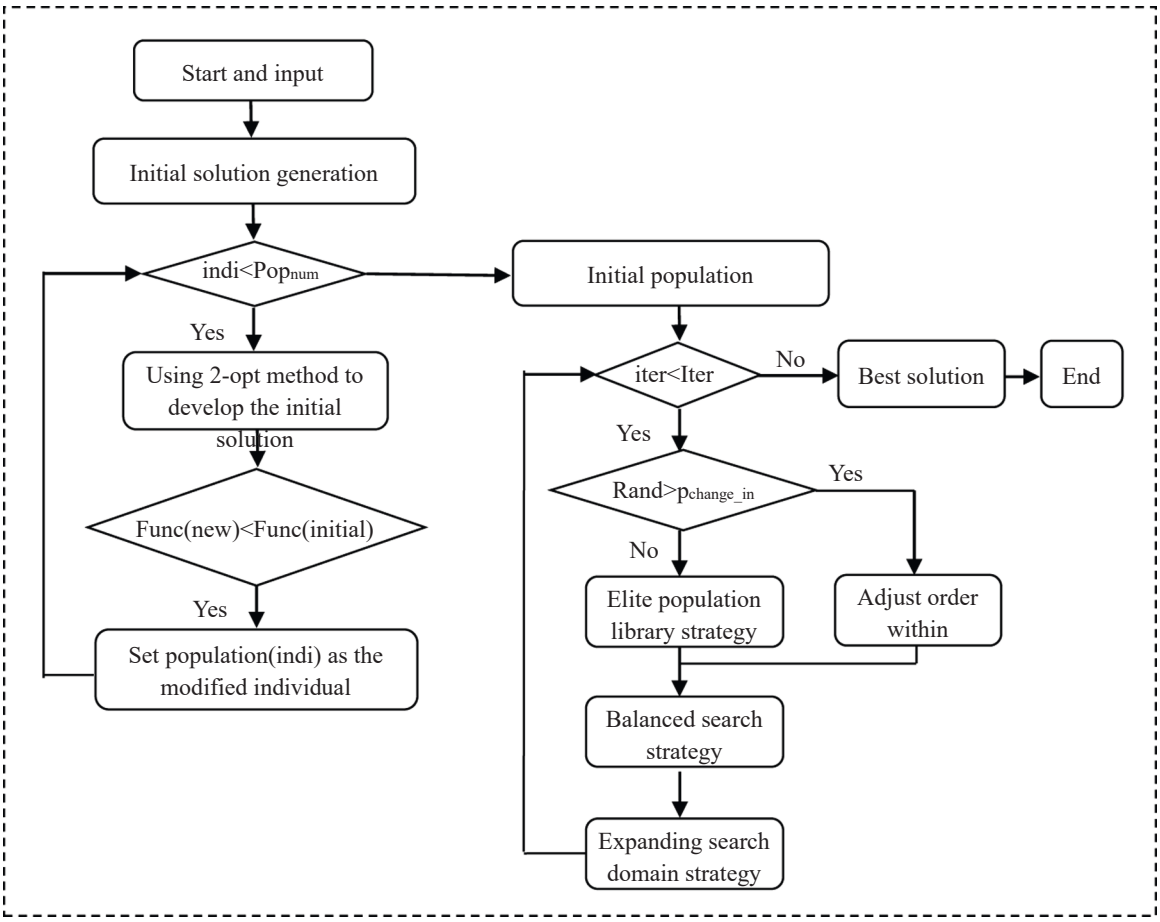


Fig. 2. Flowchart of the proposed algorithm.

lies in how to find the optimal solution under a limited number of evaluations. This study imitates the hierarchical testing mechanism that has proven effective in the COVID-19 epidemic prevention and control work and proposes a heuristic algorithm named Sa-VCO.

The COVID-19 virus pandemic has resulted in serious public health security problems worldwide spanning from 2019 to 2022 and claimed the lives of millions of people. Three distinctive features have significantly contributed to its devastating impact: its high transmissibility, rapid transmission rate, and extended incubation period. The most direct approach to prevent the spread of this virus is taking incessant test on the entire population. Then, identify infected individuals and treat them in isolation. However, due to the constraints in human resources, transportation capabilities, and available testing resources (such as nucleic acid detection kits), this method is not feasible. This predicament is similar to the challenge researchers face in large-scale combinatorial optimization problems, where evaluating all potential solutions is impractical.

To address this challenge, epidemic prevention and control personnel introduced a tiered and zoned testing measure. This involves dividing the entire population into three categories: infected individuals, close contacts, and the general public. Additionally, the areas of human activity are designated as epidemic zones and non-epidemic zones. During implementation, the entire population is initially tested, with those found to be infected labeled as such, and individuals who have had temporal or spatial proximity to them being identified as close contacts. Following this, infected individuals undergo isolation and treatment, while viral strains are extracted for vaccine development. Close contacts are quarantined and subjected to multiple tests to ascertain their carrier status. The general public residing in epidemic zones undergoes relatively frequent testing, while those in non-epidemic zones are not strictly restricted in their movements and only undergo regular periodic testing. The aforementioned mechanism effectively rationally allocates resources, minimizing unnecessary expenditures of both human and material costs, while significantly enhancing the efficiency of identifying infected individuals. Based on this mechanism, this study designed the search framework for our proposed algorithm.

In this part, we will describe in detail the framework of our proposed algorithm, as well as the strategies used in each part. First, the complete block diagram will be shown in Fig. 2, and then the mechanism and strategies will be respectively introduced. Finally, the complete pseudo-code will be referred to Table 3 in section 4.3.

4.1. Population initialization

This study fully considers the structure of the solution to the multi-UAV path planning problem, which is not a simple discrete sequence but a graph structure with practical significance. Proper pre-allocation of the inspection tasks and planning appropriate inspection sequences based on the positional relationship of these tasks will help avoid massive unnecessary search costs. In this study, an adaptive hybrid strategy selection method is designed to generate the initial solution, which aims to help the algorithm start searching in the feasible region and quickly construct a high-quality initial solution. Its complete block diagram is shown in Fig. 3.

As the figure shows, this method first used the clustering method to allocate the targets to all UAVs involved in the inspection mission as evenly as possible. Then, the initial target inspection order will be adaptively generated according to the number of targets in the sub-paths. We use a target number threshold D_{max} to help with strategy selection. For the sub-paths with a small number of targets, we use the

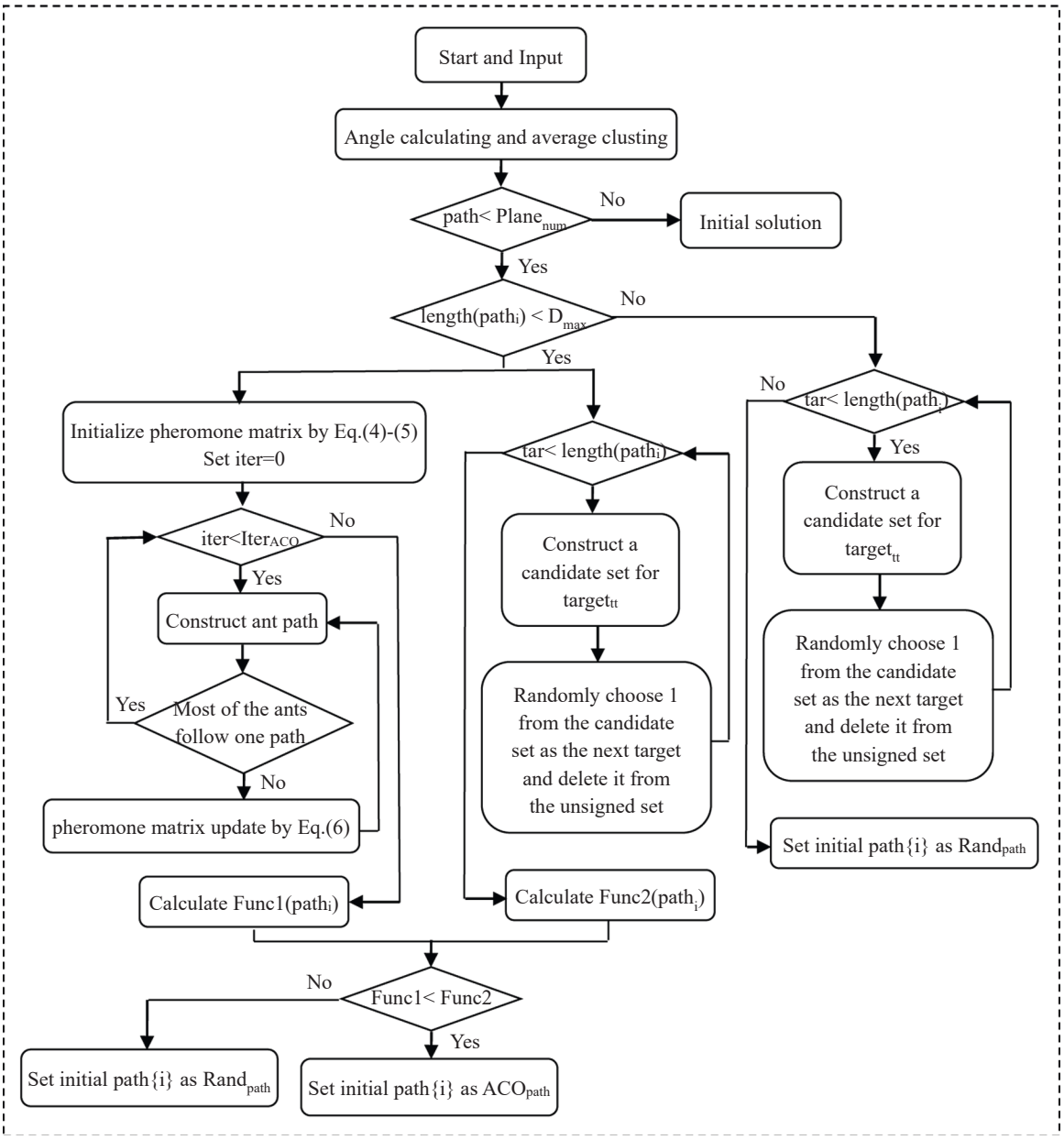


Fig. 3. Flowchart of the designed population initialization method.

ACO method and the random-greedy selection method for competitive selection at the same time. For the sub-paths with a large number of targets, we only use the random-greedy selection method to reduce the computational consumption. The mechanism and search process of each strategy will be introduce in more detail in 4.1.1 – 4.1.3.

4.1.1. P-clustering method

In this population initialization method, an improved Polar coordinate-based clustering approach is designed to pre-allocate the targets to different UAVs. Here, the distance between each target to the starting depot and the angle of the line connecting them to the starting depot relative to the horizontal positive direction are used as the clustering basis. We chose these two features because they well represent the degree to which a target is closely related to other targets in the graph structure. The clustering process consists of two steps. In step 1, a new coordinate system is first established. This coordinate system takes the starting depot as its origin, takes the horizontal direction as its x-axis direction, and takes the vertical direction as its y-axis direction. Then, the polar coordinates of all target points in the new coordinate system will be calculated as their new coordinates. In step 2, all target points will be classified into m classes (same as the number of UAVs) by their new coordinates through a set of iterative adjustments. In particular, this clustering method first sorts all targets according to their polar angle and assigns the same number of targets to each class. If the targets cannot be evenly assigned, the remaining targets will be classified into the last class. Then, assume a center for each class and evaluate the task volume for each class on our designed cost index. This index is designed based on the closeness of all targets in the class and the distance from the center to the starting depot. Its specific formula is shown as Eq. (9).

$$C_i = 2 \times (d_i - d_{\min}) + \sum_{j=1}^{num_i} d_{cj} i = 1, 2, \cdots, m; j = 1, 2, \cdots, num_i \tag{9}$$

Here, C_i is the cost index of class i, d_i is the distance from the center of class i to the starting depot, d_{min} is the minimum distance from the center of all classes to the starting depot. We use the first part of this formula to

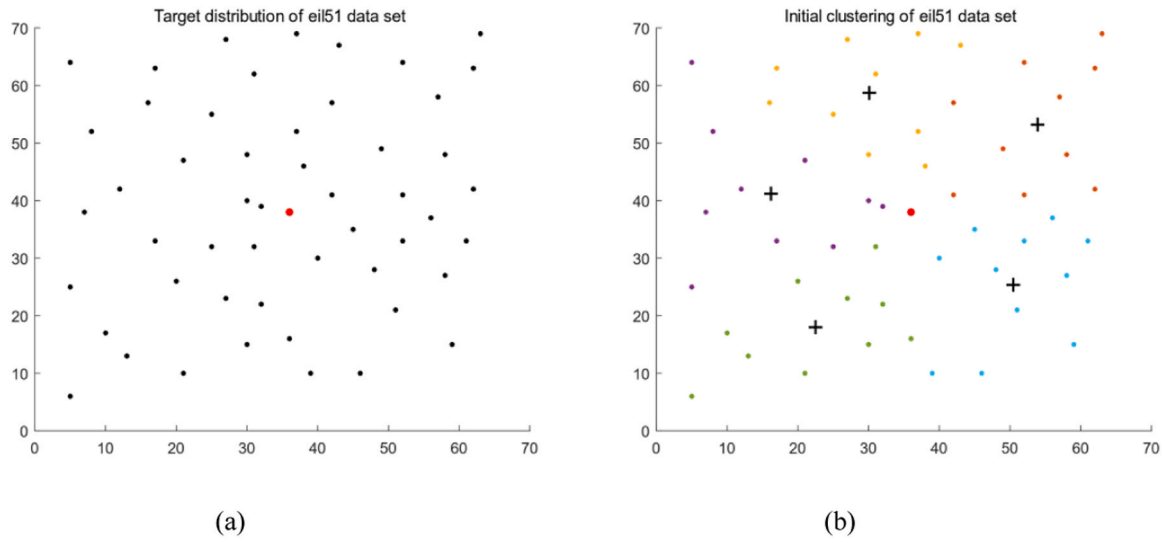


Fig. 4. Example of the P-clustering method.

Table 1
ACO Population Initialization process.

Algorithm 1 ACO Population Initialization

Input: city_position, city_number, population_number, iter_number
Output: the initial solution iniPop
1: initialize the pheromone matrix using (4);
2: initialize the choose probability matrix using (5);
3: set the colony to the start point;
4: **while** $t <$ ACO_iter **do**
5: **for** i=1 to ant_number;
6: **for** j=1 to city_number;
7: choose the next node j for ant i using the choose probability matrix;
8: **end for**
9: **end for**
10: update the pheromone matrix using (6);
11: **end while**
12: calculate the ant_pop's fitness value, choose the best as the initialpop B;

measure the difference between the path costs consumed when the UAVs flew from the starting depot to different classes for inspection tasks. d_{cj} is the distance between the target j and the center of the class it belongs to. For the second part of this formula, we measure the degree of target aggregation in each class by calculating the sum of the distance from the center of each class to all targets contained in this class. Fig. 4 shows an example of the Polar coordinate-based clustering approach. Fig. 4(a) is the target distribution of the eil51 data set and (b) gives the initial clustering result obtained by the proposed method. In both sub-figures, the red point marks the position of the starting depot. The black plus signs in (b) represent the center assumed for each class.

4.1.2. ACO initialization method

ACO is a classical swarm intelligence optimization algorithm inspired by the collaborative behavior of ant colonies in finding food. In this algorithm, a colony of ants collaborates by leaving pheromones on the paths they pass. After a period of search, the pheromone concentration on the path passed by the most ants will far exceed others, and this path is considered to be the optimal one from the nest to the food. After the optimal path is confirmed, all ants will forage along it. The ACO algorithm establishes a pheromone matrix by simulating the collaborative behavior of ant colonies. This unique component achieves a clever information positive feedback mechanism which provides strong robustness to ACO while searching for the optimal solution. The pseudo-code of the ACO algorithm is shown in Table 1.

Due to its specific advantages, many researchers choose ACO or approximate ACO methods to generate the initial population instead of generating the initial population randomly. ACO is suitable for most distributions of the data sets. However, a significant drawback is that as the data set size increases, the cost of updating the pheromone matrix becomes expensive, resulting in increased time consumption. This drawback conflicts with the trend of practical applications towards

Table 2
Random select initialization method.

Algorithm 2 Random select initialization method

Input: target_position, target_num, population_size, path_num, iter_num, start_position, class
Output: the initial solution iniPop
1: **for** i=1 to *path_num*;
2: **while** $length_path_i < class_i_number$
3: Initialize the neighbor matrix N as empty;
5: **for** j=1 to *candidate_size*;
6: choose the nearest target to the start_position in $class_i$ and add it to N;
7: delete the target from $class_i$;
8: **end for**
9: Randomly select a target from N and add the other targets in N to $class_i$;
9: Add the select target to the end of $path_i$;
10: **end while**
11: **end for**

larger data sets. In this study, we aim to leverage ACO's advantages while avoiding its limitations as much as possible. Therefore, the proposed algorithm uses ACO as the initialization method for the sub-paths with a small number of targets to start on a better set of initial solutions, thus reducing computational consumption. To avoid the expensive time costs associated with the pheromone matrix's update, the ACO method is used only once throughout the process.

Similar to regular ACO, the pheromone matrix must be initialized before the algorithm runs. In this study, the position of the depot node, ants' amount, tours amount, city nodes amount and positions are designated as inputs, and the pheromone matrix is generated using the following Eq. (10):

$$\tau_{ij} = \frac{1}{D_{ij}} \quad i,j = 1,2,\cdots,city_num, city_num+1 \tag{10}$$

Where D_{ij} represents the Euclidean distance from node i to node j. Based on it, the choose probability matrix P will be continue to generate by (11):

$$P_{ij} = \frac{\tau_{ij}^{\alpha}}{\sum_{j=1}^{n} \tau_{ij}^{\alpha}} \quad i = 1,2,\cdots,city_num, city_num+1 \tag{11}$$

Where P_{ij} represents the probability that the ant will choose node j to be the next node of node i, and α is a preset constant. In this study, we set α as 1.5 uniformly.

After that, the pheromone matrix will be updated following (12) below, with a constant of evaporation rate ρwhich is set to 0.1:

$$\tau_{ij} = (1-\rho) \times \tau_{ij} + \sum_{t=1}^{m} \Delta\tau_{ij} \tag{12}$$

4.1.3. Random-greedy select initialization method

Random strategy is an efficient method used by most intelligent optimization algorithms to generate the initial population. Unfortunately, when the number of targets to be planned increases sharply, the solution generated by this method will be quite poor. This is because the typical random method ignored the graph structure relationship between targets and may lead to a large number of impossible path segments. As a result, using the typical random method to generate the initial population will bring a great computational burden. To overcome this limitation, a random-greedy selection method is designed. In its search process, there exist two target sets: inspected target set and uninspected target set. All targets assigned to the sub-path will be first added to the uninspected target set and the inspected target set is empty at this time. Then, the entire inspection order of this sub path will be completed by multiple constrained random selections. For each selection, a candidate set with a specified number of targets will be established at first. Here, all candidates are the targets that are closest to the last target of the inspected target set and are still in the uninspected target set. If the inspected target set is empty, the start depot will be chosen as the 'last target'. Then, a target will be randomly selected from the candidate set to join the path list, add to the inspected target set, and delete from the uninspected target set. The above operations will be repeated until all the targets contained in the sub-path are added to the inspected target set. Table 2 shows the pseudo-code of our proposed method and Fig. 5 gives an example of the initial visit order provided by our designed random greedy approach.

4.2. The operational mechanism and search framework for Sa-VCO

To achieve higher search efficiency, after the initial population is generated, the complete population will be partitioned into three subpopulations: the elite subpopulation, the conventional subpopulation, and the poor subpopulation. The individuals in the elite subpopulation possess the highest quality, thus rendering the utmost information value for searching for the global optimal solution. The conventional subpopulation includes those individuals that are close in distance to the elite individuals but do not possess a similarly high solution quality. We leverage the cooperation between this subpopulation and the elite subpopulation to explore the search space, effectively partitioning it into regions of superior and inferior search potential. In particular, for the path planning problem, we compare the differences between individuals in this subpopulation and those in the elite subpopulation. By identifying the path segments that contribute to the inferior quality of these individuals, we aim to avoid incorporating such segments in our planning, thereby reducing ineffective searches. Finally, the poor subpopulation is composed of the remaining individuals, they lack significant quality and are situated relatively far from the current optimal solution. Individuals in this subpopulation are expected to explore the solution

Table 3
The framework of the proposed algorithm.

Algorithm 3 Sa-VCO Algorithm

Input: Initial population, initial func, p_d, p_b, and the input of the initial part
Output: the best solution P_{best}
1: **while** $t <$ iter_number **do**
2: sort the population by fitness value and stored P_{best};
3: divide the population into P_e, P_c, P_p;
4: **if** rand < p_{change_in}
5: **for** i=1 to *elite_number*;
6: generate excellent offspring by elite information database strategy;
7: **end for**
8: **else**
9: **for** i=1 to *elite _number*;
10: generate excellent offspring by adjusting the order within visits;
11: **end for**
12: **end if**
13: **for** i=1 to *conventional _number*;
14: randomly choose a learning individual from the excellent subpopulation;
15: generate the conventional offspring;
16: update the superior path segment memory set and the inferior path segment memory set;
17: **end for**
18: **for** i=1 to *poor _number*;
19: randomly choose a collaborative individual from the poor subpopulation;
20: generate the poor offspring;
21: improved the poor offspring based on the memory sets and record the success times;
22: **end for**
23: **end while**
24: finally select the best individual P_{best};

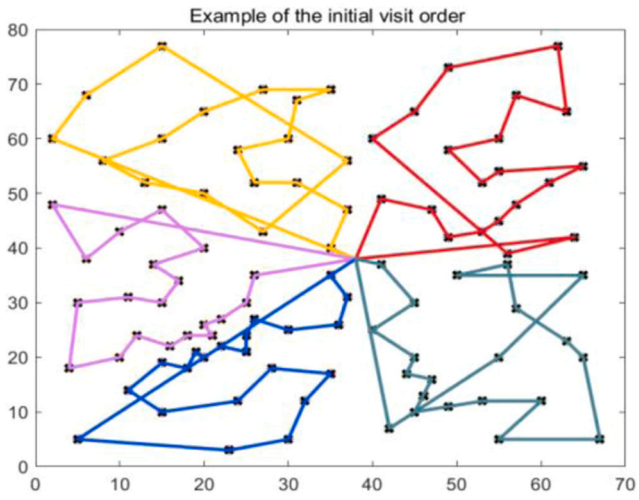


Fig. 5. Example of the initial visit order provided by the random greedy approach.

space more extensively, thereby maintaining the diversity of the population and mitigating the risk of falling into local optima.

The subdivision of the population is conducted based on the individual's fitness values, the similarity between each individual to the current optimal solution, and the subpopulation size control parameters p_d and p_b. Here, the similarity degree between any individual and the current optimal individual can be assessed by their relative distance according to Eq. (13).

$$S_j = \sum_{i=1}^{UAV_{num}} \frac{\sum_{k=1}^{U_{num}} S_{best}^{ijk}}{num_j^i} j = 1, 2, \cdots, population_{num}; i = 1, 2 \cdots, UAV_{num}; k = 1, 2 \cdots, U_{num} \quad (13)$$

In Eq. (13), S_j is the similarity degree between $individual_j$ and the current best individual, num_j^i is the number of the allocated targets in the i_{th} path of solution j, S_{best}^{ijk} is the similarity of the k_{th} target that overlapping between the i_{th} sub-path of $individual_j$ and the corresponding sub-path of the current optimal solution. It can be computed by Eq. (14). In this equation, U_{best}^{ijk} is the number of path segments that take the k_{th} target in the i_{th} path of the j_{th} individual as the endpoint and have a

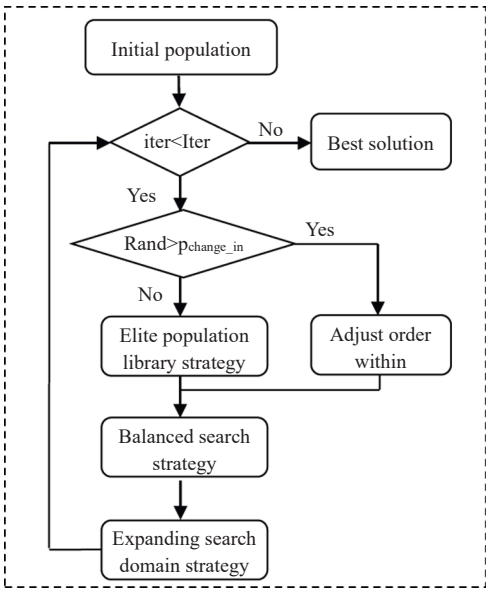


Fig. 6. Sa-VCO search framework.

consistent order relationship to a path segment in the i_{th} path of the current best individual.

$$S_{best}^{ijk} = \frac{U_{best}^{ijk}}{num_{best}^{i}} \tag{14}$$

After determining the similarity degree between each individual and the current optimal, all individuals within the population excluding the current optimal will be organized in ascending order based on their fitness and similarity values respectively. The common size (N_{ec}) of the elite subpopulation and the conventional subpopulation can be calculated by multiplying the population size and parameter p_d, and the first N_{ec} individuals in the similarity order will be divided into the assumed elite subpopulation. Then, a "cutoff individual" is defined to distinguish the individuals with relatively lower quality from the assumed elite subpopulation. We first choose the f_{th} individual in the quality order of the population. The value of f is determined by multiplying the population size and the parameter p_b. The individuals that have been allocated to the assumed elite subpopulation but have a worse fitness value than the "cutoff individual" will be taken out of the assumed elite subpopulation to form the conventional subpopulation. At this time, the remaining individuals in the assumed elite subpopulation will constitute the formal elite subpopulation.

Since the distribution of individuals in the population is likely to change at different stages, with the population mostly moving towards the area close to the optimal solution towards the end of the iteration, the acceptance probability of subpopulations will be dynamically adjusted during the algorithm's execution. The update of control parameters is performed according to Eq. (15) and Eq. (16). The complete search framework of SA-VCO is shown in Fig. 6.

$$p_d = p_{binitial} + 0.4 \times \left(\frac{1}{1 + e^{-\frac{6 \times iter}{Iter}}} - 0.5 \right) \tag{15}$$

$$p_b = p_{binitial} + 0.4 \times \left(\frac{1}{1 + e^{-\frac{6 \times iter}{Iter}}} - 0.5 \right) \times \frac{l_e}{l_e + l_c} \tag{16}$$

4.2.1. Elite population library strategy

The elite individuals share complete information with all individuals in the elite subpopulation and an elite information database is generated as the specific operation. This is done by selecting visit i with the minimum consumption in the set of each excellent individual's visit i. After removing duplicate nodes, all m selected visit i are set as the elite information base. Based on this elite information database, the elite offspring can be generated. The algorithm implements this idea by deleting the nodes contained in the elite information base from all elite individuals, and then splicing the remaining parts with the elite information base respectively. To illustrate this operation more clearly, the complete process is shown in Fig. 7.

Before the final generation of offspring, an additional refinement step utilizing both the memory set of superior path segments and the memory set of inferior path segments is introduced to further adjust the obtained offspring individuals. Specifically, this involves minimizing the overlap of segments in the offspring individuals with the inferior path segment set by deleting as many such segments as possible. Subsequently, the remaining discontinuous segments in the offspring individuals are connected using path segments from the memory set of superior path segments, whenever feasible. Lastly, random methods are employed to complete the offspring individuals into feasible solutions, ensuring that all offspring individuals meet the necessary constraints. To improve Sa-VCO's search efficiency and avoid the probability of jumping around the optimal solution without reaching the end of the iteration, the above strategy may be canceled during the iteration process. The control parameter for this is given by (17). If the cancel probability is met, the elite population will perform another strategy. In this case, the nodes contained in the current visits will be considered optimal, and only the order within each visit will be adjusted.

$$P_{change_in} = \left(\frac{iter}{Iter} \right)^{\beta} \tag{17}$$

4.2.2. Balanced search strategy

This study designed an equilibrium search strategy to leverage the disparity between conventional individuals and elite individuals for analyzing and partitioning the search space. Specifically, two memory bank sets are initially established: one for superior path segments and another for inferior path segments. During the first iteration of the search process, the path segments that are most frequently repeated among the elite individuals are recorded in the superior path segment memory set, while those most frequently repeated among the conventional individuals are recorded in the inferior path segment memory set. Subsequently, considering that our purpose is to find the path segments that are the reason for the quality difference between elite individuals and conventional individuals, any overlapping path segments between these two memory bank sets are purged, ensuring that each bank set retains unique and distinct path segments.

After establishing the superior memory bank sets and the inferior memory bank sets, a learning mechanism is introduced for the conventional offspring generation. For each individual in the conventional subpopulation, an elite individual is randomly selected from the elite subpopulations for learning. In the learning process, m visits with lower costs between the chosen elite individual and the parent conventional individual are first selected to establish a rough offspring structure. Then, the duplicate nodes in this structure will be removed and the targets that are not included will be used to construct a residual target set. All targets in the residual set will be randomly added to the rough offspring structure. Similar to the elite offspring generation process, the above two memory banks will also be applied before the final offspring is generated. This learning mechanism will improve the searchability of the conventional subpopulations by preventing Sa-VCO from searching for the poor search domain. In addition, the two memory bank sets will be updated in subsequent iterations based on the number of successful improvements to the obtained solutions.

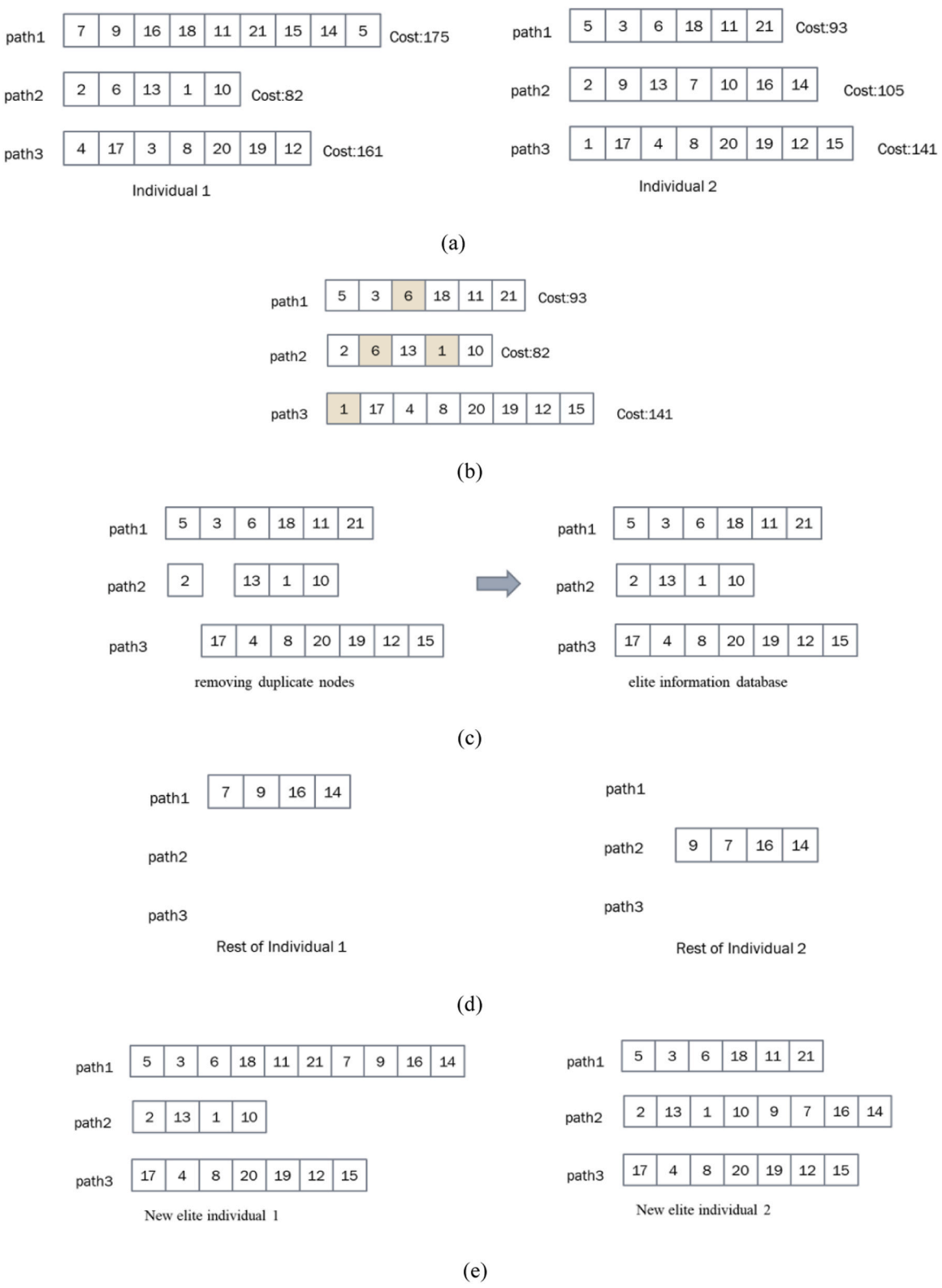


Fig. 7. Example for elite offspring generation.

4.2.3. Expanding search domain strategy

Finally, for the poor subpopulation, it is assumed that the nodes setting within each visit in the individual's solutions are not optimal, so adjusting the nodes may lead to better solutions rather than adjusting the order within the visit. Additionally, to prevent premature convergence due to a lack of diversity in the population, which may result from the elite information base strategy, the poor subpopulation will not learn from the excellent or conventional subpopulations. Instead, it will randomly select another poor individual to collaborate with and attempt to generate a better solution. The process for generating offspring is similar to that of conventional offspring generation. Although the probability of generating a better solution through this strategy may be

Table 4
Parameter settings.

parameter	value
Iter_num	400/800/4000
Pop_num	100
ACO_Iter	100
ant_num	50
ACO_r	0.1
$P_{d_initial}$	0.4
$P_{b_initial}$	0.4
D_{max}	1000
$candidate_{num}$	1/2/3/5/10

relatively weak, the 2-opt method will be applied to avoid missing a potentially better solution.

4.3. Complete process of Sa-VCO

According to the description and analysis of our proposed algorithm's initialization and update strategy above, the complete procedure of this algorithm can be summarized as follows.

Step 1: Initialize algorithm control parameters and test data set settings.

The final objective of the optimization is to minimize the max path cost of the complete planning and obtain a complete plan with m paths containing each point in the test set. In addition, the initial value of the control parameters p_d and p_b, maximum number of iteration Max_Iter and the maximum number of ACO iteration ACO_Iter are also specified in this step.

Step 2: Initialize population.

The initial population is generated in this step by the improved ACO method or the improved P_clustering method according to the test set's scale. After generation, the population is divided into three subpopulations: elite subpopulation E, conventional subpopulation C, and poor subpopulation P according to their fitness value and the similarity degree between these individuals and the current optimal.

Step 3: Update the elite information database.

The elite information database will be generated by allowing the free sharing of information between all elite individuals as described in 3.2.1.

Step 4: Update the population.

All individuals will be updated by applying the corresponding operation to each subpopulation. The 2-opt algorithm will be used to optimize all paths before calculating and updating fitness values.

Step 5: Judge the iteration end condition and output the result

If the number of iterations has reached the set value, then the optimal results are output. Otherwise, continue iterating.

5. Experimental results and analysis

To prove the superiority of the proposed algorithm, we conducted comparative tests and analyzed the results. The evaluation experiments were carried out on a personal computer with 11th Core i7, Windows 11, Matlab 2021, and the parameter settings are shown in Table 4. The number of iterations was set to 400 in the first and final part of the comparative experiment, 800 in the second and fourth parts, and 4000 in the third part of the large-scale data set test. Other algorithms involved in comparative experiments including the CPLEX, LKH-3, OR-tools, NCE, NCE-mTSP [27], HGA [28] in part one, MASVND [29], ES [30], HSNR [31], ITSHA [32], HGA [28], MA [33], AC-ABC [34] in part two, and ABC [35], IWO [35], GVNS [36], MASVND [29] in part three. These algorithms are all the current state-of-the-art algorithms. The parameter settings of all comparison algorithms are the same as those provided by the authors.

The complete experiment is divided into five parts. First, we tested the algorithm on small-scale data sets to verify its basic effectiveness and observe the cost uniformity between each path. Then, we tested its performance on multiple middle-scale data sets with different distributions to prove Sa-VCO's balance and universal applicability. Considering the algorithm's performance on large-scale data sets determines whether this algorithm has sufficient application potential, the third part of our experimentation utilizes three large-scale datasets to evaluate its efficacy comprehensively. In the fourth part, we focus on the impact of varying the candidate set size on the algorithm's performance and conduct a parameter sensitivity analysis. In the final part, an impact analysis is used to illustrate the effectiveness of our proposed improvements.

5.1. Comparative result on small-scale datasets

In the first part of the experiment, we conducted tests on four small-scale data sets with different distributions, namely eil51, berlin52, eil76, and rat99. The number of paths was set to 2, 3,5,7. The comparative results are presented in Table 5 and Table 6, and the BKS reported in this Table is the best-known solution to these instances. We further utilize Fig. 8 to visually demonstrate algorithm performance disparities more intuitively. Here, (a) is the result of eil51, (b) is the result of berlin52, (c) is eil76, and (d) is rat99. In Fig. 8, the x-axis represents the comparison algorithms in the sequence of CPLEX, LKH3, OR-Tools, NCE, NCE-mtsp, HGA, BKS, and Sa-VCO. The y-axis represents different test batches, in test 1 the path number is set to 2, test 2 is 3, test 3 is 5, and test 4 is 7. The z-axis is the gap between the best result found by these algorithms and the current optimal solution in each batch test.

The comparison results demonstrate that Sa-VCO performs well in most tests, especially those with more paths or more targets. In the tests

Table 5
Comparison results on eil51 and berlin52.

Instance	UAV_{number}	CPLEX	LKH3	OR-Tools	NCE	NCE-mtsp	HGA	BKS	Sa-VCO
eil51	2	222.7	222.7	243.3	235	226.1	222.9	222.7	**219.1**
	rank	2	2	7	6	5	4		1
	3	**159.6**	**159.6**	170.5	166.3	**159.6**	**159.6**	**159.6**	166.7
	rank	1	1	7	5	1	1		6
	5	124	124	127.5	121.6	119.9	118.3	118.3	**110.3**
	rank	5	5	7	4	3	2		1
	7	112.1	112.1	112.1	112.1	112.1	112.1	112.1	**98.4**
	rank	2	2	2	2	2	2		1
berlin52	2	4110.2	4110.2	4665.5	**4110**	4128	4110.2	**4110**	4239.1
	rank	2	2	7	1	5	2		6
	3	3244.4	3244.4	3311.3	3274	3191	**3073.8**	**3073.8**	3150.5
	rank	4	4	7	6	3	1		2
	5	2441.4	2441.4	2482.6	2660	2474	2440.9	2440.9	**2313.2**
	rank	3	3	6	7	5	2		1
	7	2440.9	2440.9	2440.9	2441	2441	2440.9	2440.9	**2188.8**
	rank	2	2	2	6	6	2		1

Table 6
Comparison results on eil76 and rat99.

Instance	UAV $_{number}$	CPLEX	LKH3	OR-Tools	NCE	NCE-mtsp	HGA	BKS	Sa-VCO
eil76	2	**280.9**	**280.9**	318	285.5	282.1	281.6	**280.9**	**280.9**
	rank	1	1	7	6	5	4		1
	3	197.3	197.3	212.4	211	197.5	**196.7**	**196.7**	201.3
	rank	2	2	7	6	4	1		5
	5	150.3	150.3	143.4	144.6	147.2	143.4	143.4	**135.5**
	rank	6	6	2	4	5	2		1
	7	139.6	139.6	128.3	127.6	127.6	127.6	127.6	**107.2**
	rank	6	6	5	2	2	2		1
rat99	2	728.8	728.8	762.2	695.8	666	668	666	**656.3**
	rank	5	5	7	4	2	3		1
	3	587.2	587.2	552.1	527.8	533.2	517.8	517.8	**450.5**
	rank	6	6	5	3	4	2		1
	5	469.3	469.3	473.7	458.6	462.2	450.3	450.3	**313**
	rank	5	5	7	3	4	2		1
	7	443.9	443.9	442.5	441.6	443.9	437.3	437.3	**318.6**
	rank	5	5	4	3	5	2		1
Ave rank		3.5625	3.625	5.3125	4.25	3.8125	2.125		**1.9375**
Total rank		3	4	7	6	5	2		1

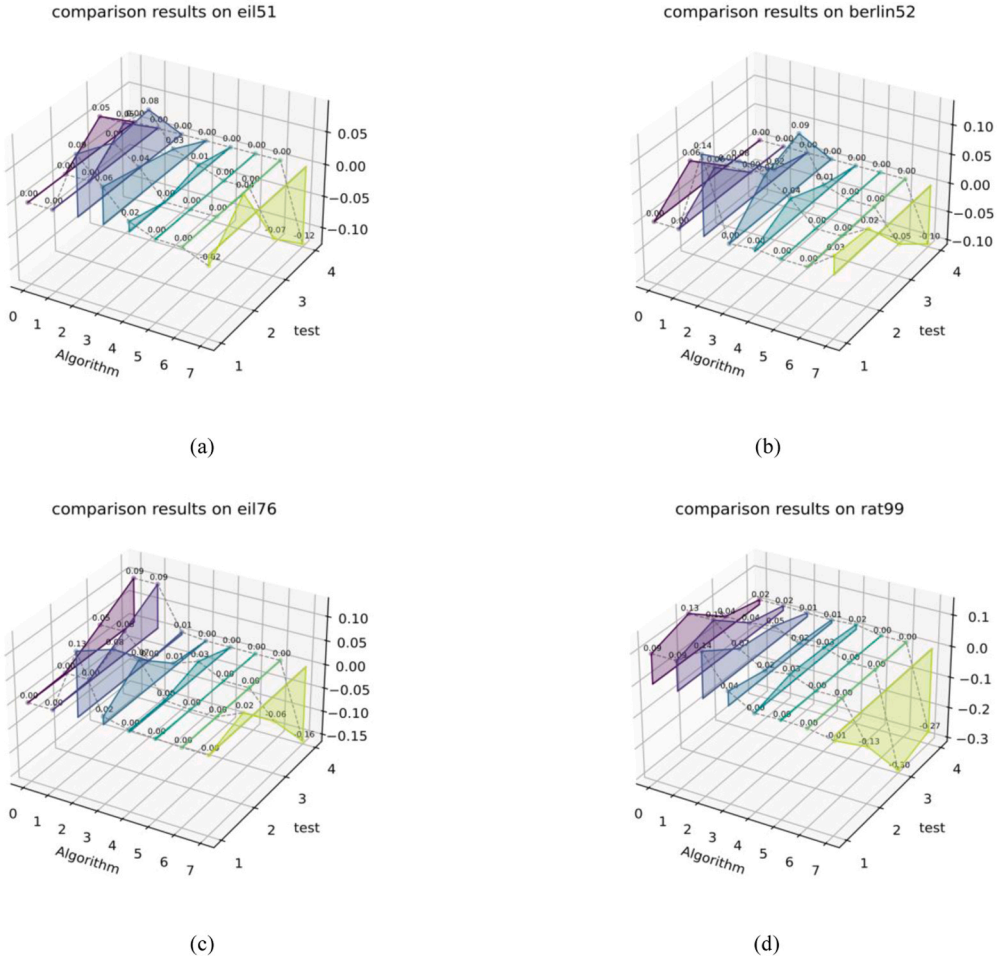


Fig. 8. The comparison results on eil51, berlin52, eil76, rat99.

on eil51, despite a 4 % gap from the previous BKS when the path number is set to 3, the Sa-VCO demonstrates a superior performance by achieving 2 %, 7 %, and 12 % advantages over the BKS when the path number is set to 2, 5, and 7, respectively. we can also see that when the number of paths and targets are both large, for example, in rat99 's tests, Sa-VCO surpasses all other comparison algorithms. This is attributed to the ingenious initial sub-path target allocation strategy employed in Sa-VCO, which enables the algorithm to construct a more optimal initial search population. Consequently, the algorithm can circumvent a significant number of ineffective searches and achieve planning results with more balanced sub-path costs.

5.2. *Comparative result on multi-type datasets*

In the second part of the experiment, we tested the performance of the proposed algorithm on 6 middle-size datasets with varying

Table 7
Comparison results on 5 middle-size data sets.

Instance	UAVnumber	MASVND	ES	HSNR	ITSHA	HGA	MA	AC-ABC	BKS	Sa-VCO
ch150	5	1758.08	1741.61	1744.26	1752.39	1741.82	1741.61	1740.556	1740.556	**1686.481**
	rank	8	3	6	7	5	3	2	0	1
	10	1554.64	1554.64	1554.64	1554.33	1554.64	1554.64	1291.01	1291.01	**1107.857**
	rank	4	4	4	3	4	4	2	0	1
	20	1554.64	1554.64	1554.64	1554.33	1554.64	1554.64	**1007.863**	1007.863	1019.746
	rank	4	4	4	3	4	4	1	0	2
kroa200	5	7415.54	7536.91	7536.91	7418.87	7449.22	7415.54	8600.241	7412.12	**7020.45**
	rank	2	6	6	4	5	2	8	0	1
	10	6223.22	6223.22	6223.22	6223.22	6223.22	6223.22	6378.624	6223.22	**4883.984**
	rank	2	2	2	2	2	2	8	0	1
	20	6223.22	6223.22	6223.22	6223.22	6223.22	6223.22	4648.39	4648.39	**4239.839**
	rank	3	3	3	3	3	3	2	0	1
lin318	5	11572.41	11604.2	11458.2	11811.14	11289.26	11295.2	10559.86	10559.86	**10195.14**
	rank	6	7	5	8	3	4	2	0	1
	10	9731.17	9731.17	9731.17	9731.17	9731.17	9731.17	8416.676	8416.676	**6108.505**
	rank	3	3	3	3	3	3	2	0	1
	20	9731.17	9731.17	9731.17	9731.17	9731.17	9731.17	7226.226	7226.226	**5046.735**
	rank	3	3	3	3	3	3	2	0	1
att532	5	22619.66	23089.7	28945	22876.24	22266.07	27067	29629.08	22266.07	**21410.69**
	rank	3	5	7	4	2	6	8	0	1
	10	18390.46	18059.7	31038.8	18091.15	18042.64	**15709**	18816.19	15709	16763.09
	rank	6	4	8	5	3	1	7	0	2
	20	17641.16	17641.2	33696.65	17641.16	17641.16	15580	14914.46	14914.46	**11569.28**
	rank	4	7	8	4	4	3	2	0	1
rat783	5	2092.77	2127.99	2066.38	2024.27	**1996.33**	2006.46	2527.175	1961.12	2087.455
	rank	6	7	4	3	1	2	8	0	5
	10	1432.34	1360.89	1358.06	1367.98	1320.35	1334.76	1576.752	1313.01	**1162.127**
	rank	7	5	4	6	2	3	8	0	1
	20	1260.88	1231.69	1231.69	1235	1231.72	1231.69	1161.701	1161.701	**929.8612**
	rank	8	3	3	7	6	3	2	0	1

distributions. The dataset sizes ranged from 150 to 1173, and the number of UAVs was set to 5, 10, and 20, respectively. The baseline algorithms include MASVND [29], ES [30], HSNR [31], ITSHA [32], HGA [28], MA [33], and AC-ABC [34]. The cut-off time of all algorithms is set to n/5 s. Table 7 and Table 8 recorded the optimal results obtained by all algorithms. To show the difference in algorithm performance more clearly, we use Fig. 9. to express the difference between the optimal results of all algorithms and the previous BKS. Further, we provide part of the path results obtained by Sa-VCO in Fig. 10 to show the optimization effect of the algorithm more intuitively. Here (a) is the result of ch150, (b) is kroA200, (c) is lin318, (d) is att532, (e) is rat783, and (f) is pcb1173.

The comparison clearly shows that Sa-VCO produces optimal and well-balanced path solutions, further demonstrating its superiority over the other algorithms. Upon observing the results, we found that the Sa-VCO exhibited a more pronounced advantage in the tests on medium-sized datasets compared to those on small-scale datasets. This can be attributed to the fact that all the comparison algorithms possess the capability to plan high-quality paths when the number of targets is relatively less. However, as the number of targets and paths increases, the search space becomes increasingly complex, offering a more comprehensive testing ground for algorithm performance. In these tests, the Sa-VCO significantly demonstrated its superior search capabilities.

5.3. Comparative result on large-scale datasets

As previously stated, the escalating scale of the problem poses a paramount challenge in multi-UAV path planning. Consequently, the devised algorithm must possess the capability to accommodate large-scale datasets. In the third part of the experiment, we subjected the Sa-VCO algorithm to rigorous testing on three large-scale datasets: u2152, pcb3038, and fnl4461. For each of these datasets, we conducted three distinct trials, varying the number of paths under consideration to 5, 10, and 20, respectively. The benchmark algorithms chosen for comparative analysis encompassed ABC [35], IWO [35], GVNS [36], and MASVND [29]. The outcomes of this comparative assessment are comprehensively provided in Table 9.

As shown in the Table, the proposed algorithm still achieves good results when the dataset size reaches 4000 and performs better than all other baseline algorithms in eight of the nine sets of tests. This demonstrates that Sa-VCO does have outstanding advantages in solving large-scale multipath planning problems and the structure and strategies we designed are effective.

5.4. Parameter-sensitivity experiment

In the Sa-VCO algorithm, an adjustable parameter exists with a non-negligible impact that influences the algorithm's effectiveness: the size of the candidate target neighborhood. When this parameter is set to a larger value, the algorithm exhibits increased randomness, making it more prone to escaping local optima. However, this enhancement in randomness simultaneously leads to reduced stability, where the initial solutions generated by the random-greedy method may necessitate the algorithm to consume significantly more computational resources in its search for the optimal solution. In this section, we complement our analysis with a parameter sensitivity examination, aiming to rigorously test the impact of this parameter on the algorithm's performance characteristics.

We utilized a middle-size dataset ch150 to conduct this testing. The testing was segmented into two distinct groups, where the number of paths was set to 5 and 10, respectively. For each of these test scenarios, we further explored the performance with varying sizes of the candidate set, specifically at 1, 2, 3, 5, and 7. Within each parameter configuration, we execute 30 independent runs. All the test outcomes are recorded and comprehensively presented through Table 10 and Fig. 11, offering a clear visualization and analysis of the experimental findings. In Fig. 11, (a) is the result of 5 paths, and (b) is the result of 10 paths.

As the results demonstrate, a larger candidate set leads to enhanced diversity, enabling the algorithm to potentially explore better solutions. However, this comes at the cost of reduced stability, making it challenging to guarantee the solution quality every time, especially under constrained search resources. Conversely, a smaller candidate set

Table 8
Comparison results on pcb1173.

Instance	UAVnumber	MASVND	ES	HSNR	ITSHA	HGA	MA	AC-ABC	BKS	Sa-VCO
pcb1173	5	14557.3	14099.5	13402.3	12952.97	**12812.2**	12952.97	17038.49	12812.2	14522.86
	rank	7	5	4	2	1	2	8	0	6
	10	9222.92	8160.25	8120.45	7864.11	7681.41	7758.26	9993.305	7623.59	**7450.33**
	rank	7	6	5	4	2	3	8	0	1
	20	7063.23	6549.14	6528.86	6528.86	6528.86	6528.86	6721.947	6528.86	**5195.408**
	rank	8	6	2	2	2	2	7	0	1
Ave rank		3.791667	3.458333	3.375	3.041667	2.291667	2.208333	3.625	0	1.208333
Total rank		8	6	5	4	3	2	7	0	1

imparts greater stability to the algorithm but also increases the likelihood of becoming trapped in local optima. Nevertheless, when computational resources are abundant, the search paradigm of Sa-VCO equips it with the capability to escape from suboptimal solutions. Consequently, this parameter significantly impacts the algorithm's search capability only when computational resources are extremely limited. In practical applications, the setting of this parameter should be determined to the specific requirements and constraints.

5.5. Impact analysis experiment

The initial population generation method is a crucial component of our proposed algorithm. To demonstrate the enhancement in algorithm performance attributable to this initialization technique, the final part of the experiment conducts an impact analysis test. This test is performed across four datasets of varying sizes, which encompass 51 targets, 99 targets, 150 targets, and 532 targets, respectively. In each test, the number of drones is consistently set to 5, with the iterative search strategy and evaluation criteria remaining the same. The initial solution generation methods included in the comparison comprise the complete adaptive hybrid initialization method based on the P-clustering technology, P-clustering with a random-greedy strategy, P-clustering integrated with the ACO algorithm, P-clustering utilizing a random strategy, k-means clustering method with random, fuzzy C-means (FCM) clustering method with random, and a completely random strategy.

In the comparative test framework, we rigorously evaluate both the quality of the initial solution and the time required for its generation of the two independent strategies incorporated into our proposed hybrid initialization method: P-clustering with a random-greedy strategy and P-clustering integrated with the ACO algorithm. We also test the performance of the P-clustering method utilizing a random strategy to assess the effectiveness of the random-greedy approach in planning task sequences within initial sub-paths. In addition, the remaining three methods represent prevalent initial solution generation strategies in current related research. For each set of tests, we independently conduct 20 runs and record several key metrics: the average fitness value and variance of the initial solution generated by each method, the average time taken for initial solution generation, and the average fitness value and variance of the final optimized result based on that initial solution. The results shown in Table 11 indicate that the quality of the initial solution significantly influences the outcome, particularly in large-scale problems, when the search is conducted with an equal number of evaluations and strategies.

Upon comparing and analyzing various initial solution generation methods, we find that the P-clustering method integrated with the Ant Colony Optimization (ACO) algorithm, yields the highest quality initial solutions. However, this approach is notably more computationally intensive, with the calculation time increasing dramatically as the number of targets escalates. Additionally, we found that the initial solutions generated by P-clustering combined with a random-greedy strategy are of the second-highest quality, trailing closely behind the ACO-integrated P-clustering strategy. Importantly, this random-greedy strategy consumes comparable amounts of time to other methods. The adaptive hybrid initialization method proposed in this paper is collaboratively constructed from these two strategies. The test results, presented in the first column of Table 11, reveal that this method effectively combines the advantages of both strategies, enabling the acquisition of better initial solutions with relatively minimal time costs. Furthermore, the quality of initial solutions obtained by all three methods surpasses that of the K-means clustering method, the Fuzzy C-means (FCM) clustering method, and the completely random method.

Based on the test results and analysis presented above, it can be concluded that our population initialization method, which integrates the P-clustering method, ACO algorithm, and random greedy strategy, is highly effective for large-scale multi-UAV path planning problems. This approach not only generates superior initial solutions with greater

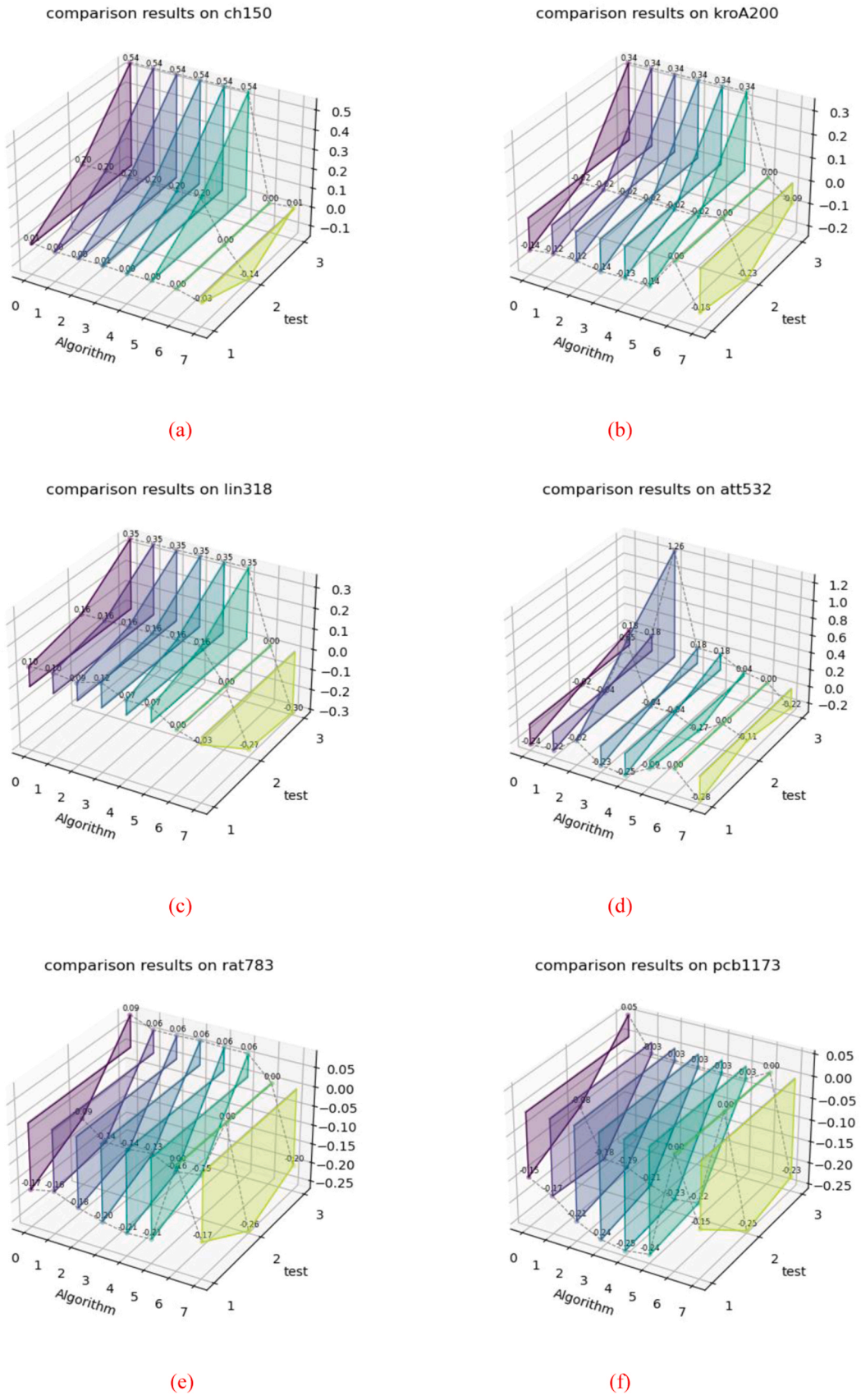


Fig. 9. The comparison results on ch150, kroa200, lin318, att532, rat782, pcb1173.

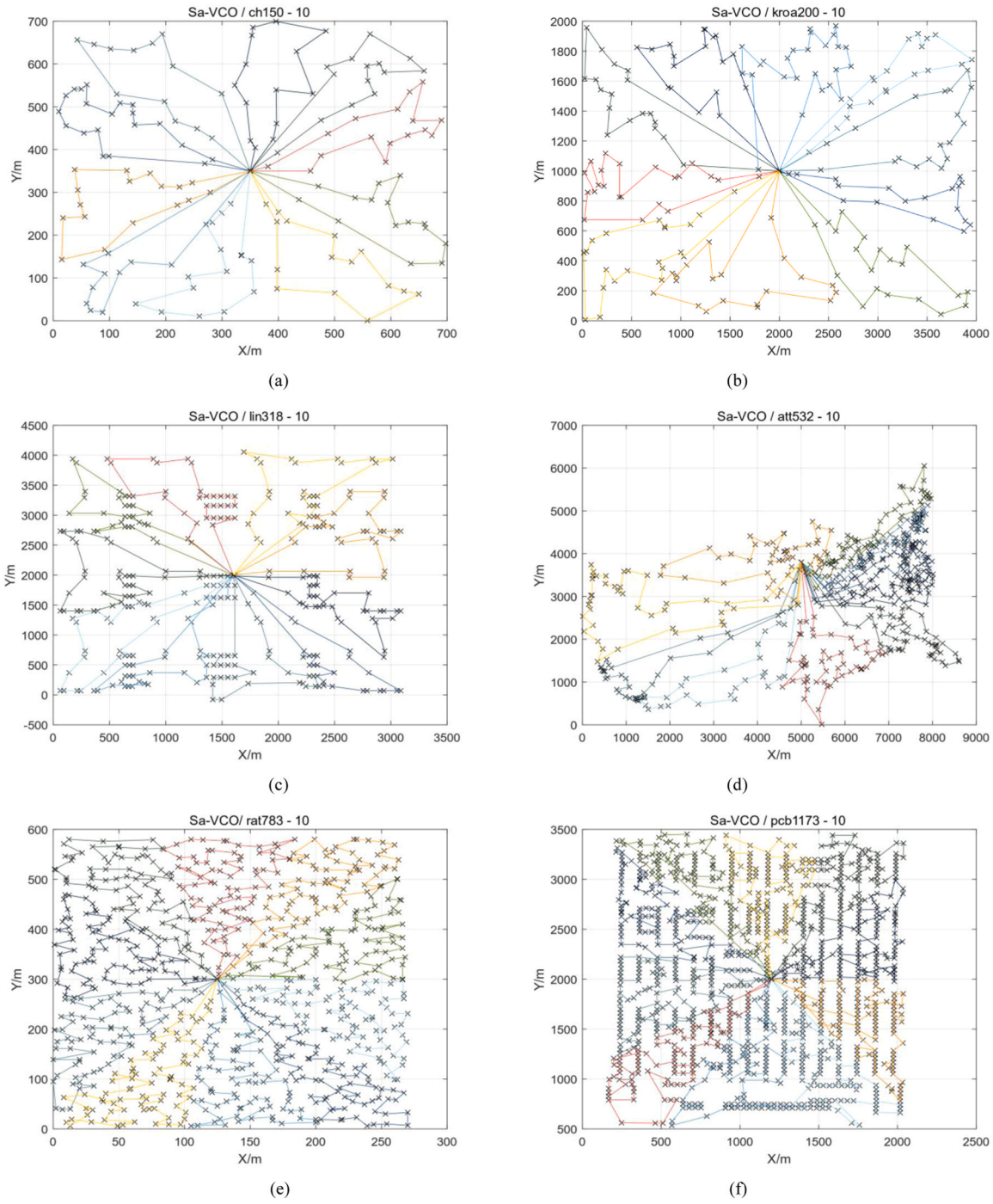


Fig. 10. The 10-path results provide by Sa-VCO on 6 middle-size data sets.

Table 9
Comparison results on 3 large-scale target sets.

Instance	UAVnumber	ABC	IWO	GVNS	MASVND	BKS	Sa-VCO
u2152	5	23571.56	23534.98	17387.25	18404.22	17387.25	**17181.95**
	10	14211.38	13985.98	10193.25	**9600.79**	**9600.79**	9722.7
	20	9609.48	9532.87	6811.78	6727.13	6727.13	**5928.528**
pcb3038	5	46858.06	46938.1	37126.47	67071.9	37126.47	**37056.36**
	10	27700.36	27659.07	20851.11	19620.41	19620.41	**19230.65**
	20	18507.84	18383.59	12955.65	12643.54	12643.54	**11406.33**
fnl4461	5	60047.22	60170.68	49343.13	84430.87	49343.13	**48928.29**
	10	34942.84	34741.06	26871.35	43581.63	26871.35	**25336.74**
	20	22798.53	22852.8	16341.6	15262.97	15262.97	**14861.89**

Table 10
The results of parameter-sensitive experiment.

Instance	path_num	result	Cn= 1	Cn= 2	Cn= 3	Cn= 5	Cn= 7
ch150	5	best	1535.533	1540.651	**1522.093**	1554.633	1534.228
		mean	**1585.492**	1586.521	1651.064	1626.493	1647.433
		worst	1669.109	**1626.127**	1816.735	1774.792	1834.631
		std	42.54781	**35.82956**	102.3517	70.65604	84.12993
	10	best	916.4731	**911.4554**	918.2077	933.5236	936.6762
		mean	**936.1634**	965.0735	984.7411	982.287	981.5403
		worst	**1021.701**	1092.273	1169.974	1062.563	1032.598
		std	33.87435	49.68766	68.94809	41.03779	**32.23014**

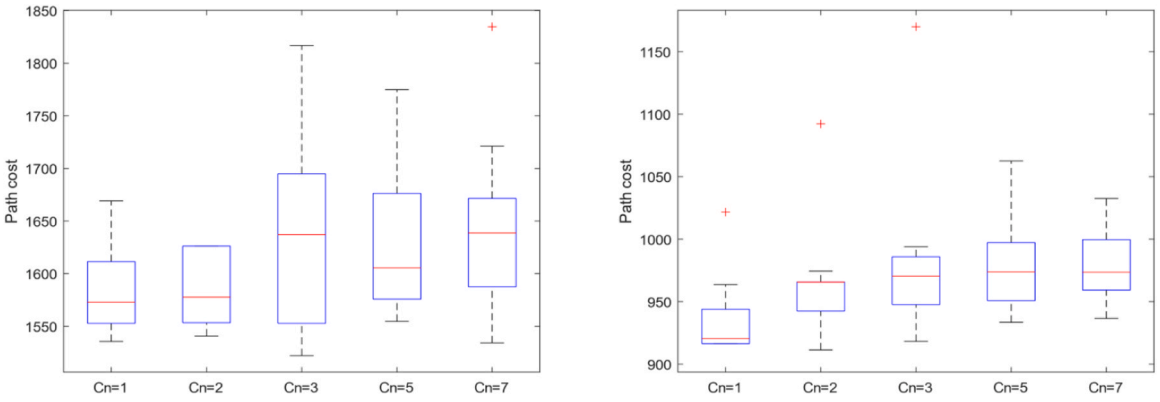


Fig. 11. The results of parameter-sensitive experiment.

Table 11
The results of the impact analysis experiment.

eil51-5		complete method	P-Greedy	P-ACO	Cluster	k-means	FCM	random
	initial_mean	**152.9495**	214.7796	153.3913	228.6861	281.7427	249.2375	418.6642
	initial_std	**2.820475**	43.63774	4.666934	2.970162	13.06512	13.7496	114.5093
	initial_time	1.560257	**0.003149**	15.27411	0.010469	0.008964	0.007838	0.006524
	result_mean	**108.8433**	113.4597	111.1848	112.5183	121.3557	123.0541	174.2483
	result_std	6.169628	12.44925	14.62005	**5.106904**	6.463191	11.01418	33.7524
rat99–5	initial_mean	525.7776	792.8027	**476.9696**	1054.305	1120.092	968.369	2098.513
	initial_std	**29.44214**	236.76	73.61592	62.99144	69.09942	66.37652	929.9325
	initial_time	1.645343	**0.004796**	15.6872	0.011944	0.011041	0.010455	0.009444
	result_mean	311.8656	336.9722	**311.195**	318.8288	351.6443	338.211	533.8015
	result_std	13.69262	47.36464	18.35277	**5.916843**	24.63036	18.90095	104.1767
ch150–5	initial_mean	2181.501	3825.519	**2114.038**	6407.951	7391.423	6642.022	12277.54
	initial_std	**144.3016**	601.023	504.3301	676.2037	746.2362	348.6915	1770.393
	initial_time	1.766813	0.014524	31.2468	0.015047	0.013926	**0.011938**	0.01204
	result_mean	1615.025	1611.087	**1535.727**	1722.949	1750.388	1773.963	2825.612
	result_std	**21.98192**	202.4174	102.06	93.7355	94.86012	88.89355	384.7722
att532–5	initial_mean	35385.33	64545.87	**29664.64**	78609.02	155007.2	151269.8	362850.3
	initial_std	1629.447	12524.92	**1237.323**	8610.677	5785.315	2219.655	52975.29
	initial_time	53.59414	0.123189	202.6961	0.187952	0.136166	**0.108851**	0.189804
	result_mean	**21872.47**	22804.37	22426.28	24790.39	38486.55	38809.8	46699.25
	result_std	897.9899	**494.3686**	1028.999	1398.973	1234.261	1381.82	1842.286

efficiency but also significantly reduces the complexity of tackling such challenging scenarios. By leveraging the combined strengths of these components, our method accelerates the problem-solving process and enhances the feasibility of managing complex situations involving numerous UAVs, thereby fulfilling our original intention of solving these problems effectively and efficiently.

6. Conclusions

In this study, we present a self-adaptive region discretization approach and an innovative heuristic algorithm, tailored to overcome limitations of existing algorithms in efficiency and quality when solving large-scale multi-UAV coverage path planning problems. By meticulously partitioning the origin target regions into a standardized grid system, our approach lets any target region be inspected by more than one UAV and significantly enhancing the flexibility and efficiency of UAV flight path optimization. At the same time, our core contribution lies in a heuristic algorithm named Sa-VCO. This algorithm includes a self-adaptive population initialization methodology and a sophisticated collaborative search mechanism that harnesses the power of multiple subpopulations. In the iterative search process, both the current quality of all individuals within the population and the degree of similarity between each individual and the current optimal solution are continuously assessed. Based on these two metrics, the entire population is

systematically partitioned into three distinct subpopulations. Subsequently, we assign specific search tasks to each subpopulation, leveraging their unique capabilities and characteristics. Through effective collaboration among these subpopulations, our algorithm achieves comprehensive analysis of the search space, evaluation of search regions, and prediction of local optimal regions. This orchestrated framework endows the algorithm with remarkable search efficiency and prowess, enabling it to explore the complex search space with precision and agility. Multiple experiments in 4 verified that the Sa-VCO has strong search ability and search efficiency. We tested our algorithm on multiple typical TSP test sets and compared it with several state-of-the-art algorithms. Our proposed algorithm showed significant advantages, particularly on large-scale data sets, making it highly applicable for multi-UAV coverage inspection on multi-regions. Overall, our proposed algorithm can adapt to the development trend of the continuous expansion of the inspection area, the increase of the inspection target, and the increasingly strict time restriction in the current multi-UAV cooperative inspection application. We believe that this method has application potential in various fields such as disaster rescue, intelligent agriculture, environment protection, and wireless sensor networks that we have mentioned above.

However, several avenues for future research exist. Firstly, the present research on multi-UAV coverage inspection on large regions is mainly based on the mTSP model and takes into less consideration the repeated access of UAVs to visited targets. In practical applications, the drones may need to repeatedly inspect some of the visited targets during the inspection process. For example, in emergency search and rescue, the UAVs will conduct a coverage inspection of all target regions and return the collected inspection data to the ground station in the first inspection. After that, the ground station will analyze the situation of the affected area based on these data, such as judging whether there are trapped people. To achieve real-time and efficient rescue work, the drone will be required to repeatedly inspect the area where there are trapped people to collect more data needed for rescue work. At this time, the problem model will be transformed into a dynamic multi-access traveling salesman model, and the difficulty of solving will be further improved. Secondly, heterogeneous models of UAVs with different structures or UAV-vehicle collaboration should also be considered. By splitting the complete inspection task into multiple steps and using different execution units to complete it, the execution efficiency of the task will be effectively improved, and the use of limited resources can be maximized.

Compliance with ethical standards

The authors declare that there is no conflict of interests regarding the publication of this paper. All authors have no conflict of interests. This article does not contain any studies with human participants performed by any of the authors.

CRediT authorship contribution statement

Chang He: Development or design of methodology, Writing-Original draft preparation and revise, Resources and simulation. **Haibin Ouyang:** Development or design of methodology, Writing-Original draft preparation and revise, Resources and simulation. **Weiqing Huang:** Algorithm development and design, revise draft and administration. **Steven Li:** Writing - Review & Editing, analysis. **Chunliang Zhang:** Review & Editing, analysis. **Weiping Ding:** Review & Editing, analysis. **Zhi-Hui Zhan:** Review & Editing, analysis.

Declaration of Competing Interest

The authors declare that there is no conflict of interests regarding the publication of this paper. All authors have no conflict of interest. This article does not contain any studies with human participants performed by any of the authors.

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Data availability

Data will be made available on request.

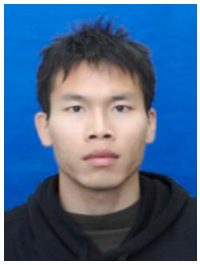
References

[1] A. Khan, S. Gupta, S.K. Gupta, Emerging UAV technology for disaster detection, mitigation, response, and preparedness, J. Field Robot. 39 (2022) 905–955, https://doi.org/10.1002/rob.22075.
[2] A. Mukherjee, S. Misra, N.S. Raghuwanshi, A survey of unmanned aerial sensing solutions in precision agriculture, J. Netw. Comput. Appl. 148 (2019) 102461, https://doi.org/10.1016/j.jnca.2019.102461.
[3] H. Tian, Z. Mo, C. Ma, et al., Design and validation of a multi-objective waypoint planning algorithm for UAV spraying in orchards based on improved ant colony algorithm, Front. Plant Sci. 14 (2023) 1101828, https://doi.org/10.3389/fpls.2023.1101828.
[4] Y. Xia, W. Liu, J. Xie, et al., Characterization of low-power wireless links in UAV-assisted wireless-sensor network, IEEE Internet Things J. 10 (2023) 5823–5842, https://doi.org/10.1109/JIOT.2022.3233576.
[5] Y. Wang, M. Chen, C. Pan, et al., Joint optimization of UAV trajectory and sensor uploading powers for UAV-assisted data collection in wireless sensor networks, IEEE Internet Things J. 9 (2022) 11214–11226, https://doi.org/10.1109/JIOT.2021.3126329.
[6] B. Zhou, H. Xu, S. Shen, RACER: rapid collaborative exploration with a decentralized multi-UAV system, IEEE Trans. Robot. 39 (2023) 1816–1835, https://doi.org/10.1109/TRO.2023.3236945.
[7] K. Li, X. Yan, Y. Han, Multi-mechanism swarm optimization for multi-UAV task assignment and path planning in transmission line inspection under multi-wind field, Appl. Soft Comput. 150 (2024) 111033, https://doi.org/10.1016/j.asoc.2023.111033.
[8] F. Ge, K. Li, W. Xu, et al., Path Planning of UAV for Oilfield Inspection Based on Improved Grey Wolf Optimization Algorithm, in: 2019 Chinese Control And Decision Conference (CCDC), IEEE, Nanchang, China, 2019, pp. 3666–3671, https://doi.org/10.1109/CCDC.2019.8833455.
[9] X. Wang, J.-S. Pan, Q. Yang, et al., Modified mayfly algorithm for UAV path planning, Drones 6 (2022) 134, https://doi.org/10.3390/drones6050134.
[10] H. Xiang, X. Liu, X. Song, et al., UAV Path Planning Based on Enhanced PSO-GA, in: L. Fang, J. Pei, G. Zhai, R. Wang (Eds.), Artificial Intelligence, Springer Nature Singapore, Singapore, 2024, pp. 271–282, https://doi.org/10.1007/978-981-99-9119-8_25.
[11] B. Han, T. Qu, X. Tong, et al., Grid-optimized UAV indoor path planning algorithms in a complex environment, Int. J. Appl. Earth Obs. Geoinf. 111 (2022) 102857, https://doi.org/10.1016/j.jag.2022.102857.
[12] H. Ali, G. Xiong, M.H. Haider, et al., Feature selection-based decision model for UAV path planning on rough terrains, Expert Syst. Appl. 232 (2023) 120713, https://doi.org/10.1016/j.eswa.2023.120713.
[13] L. Luo, X. Wang, J. Ma, et al., GrpAvoid: multigroup collision-avoidance control and optimization for UAV swarm, IEEE Trans. Cybern. 53 (2023) 1776–1789, https://doi.org/10.1109/TCYB.2021.3132044.
[14] D. Seo, J. Kang, Collision-avoided tracking control of UAV using velocity-adaptive 3D local path planning, Int. J. Control, Autom. Syst. 21 (2023) 231–243, https://doi.org/10.1007/s12555-021-0666-z.
[15] W. Zhang, C. Peng, Y. Yuan, et al., A novel multi-objective evolutionary algorithm with a two-fold constraint-handling mechanism for multiple UAV path planning, Expert Syst. Appl. 238 (2024) 121862, https://doi.org/10.1016/j.eswa.2023.121862.
[16] X. Zhao, R. Yang, L. Zhong, et al., Multi-UAV path planning and following based on multi-agent reinforcement learning, Drones 8 (2024) 18, https://doi.org/10.3390/drones8010018.
[17] P. Mahajan, A. Kumar, G.S.S. Chalapathi, et al., EFTA: an energy-efficient, fault-tolerant, and area-optimized uav placement scheme for search operations. in: IEEE INFOCOM 2022 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), IEEE, New York, NY, USA, 2022, pp. 1–6, https://doi.org/10.1109/INFOCOMWKSHPS54753.2022.9798305.
[18] R.M. Azadur, C.J. Pawase, K. Chang, Multi-UAV path planning utilizing the PGA algorithm for terrestrial IoT sensor network under ISAC framework, Trans. Emerg. Telecommun. Technol. 35 (2024) e4916, https://doi.org/10.1002/ett.4916.
[19] Y.-Z. Zhang, L. Chen, Multi-UAVs Cooperative Reconnaissance Based on Improved Harmony Search Algorithm, in: 2018 5th International Conference on Information

Science and Control Engineering (ICISCE, IEEE, Zhengzhou, 2018, pp. 346–350, https://doi.org/10.1109/ICISCE.2018.00079.
[20] L. Cheng, L. Zhong, S. Tian, et al., Task assignment algorithm for road patrol by multiple UAVs with multiple bases and rechargeable endurance, IEEE Access 7 (2019) 144381–144397, https://doi.org/10.1109/ACCESS.2019.2944881.
[21] Y. Cui, W. Dong, D. Hu, et al., The application of improved harmony search algorithm to multi-UAV task assignment, Electronics 11 (2022) 1171, https://doi.org/10.3390/electronics11081171.
[22] L. Huo, J. Zhu, G. Wu, et al., A novel simulated annealing based strategy for balanced UAV task assignment and path planning, Sensors 20 (2020) 4769, https://doi.org/10.3390/s20174769.
[23] J. Liu, Y. Zhang, X. Wang, et al., Min-max Path Planning of Multiple UAVs for Autonomous Inspection, in: 2020 International Conference on Wireless Communications and Signal Processing (WCSP), IEEE, 2020, pp. 1058–1063, https://doi.org/10.1109/WCSP49889.2020.9299869.
[24] X. Yan, R. Chen, Z. Jiang, UAV cluster mission planning strategy for area coverage tasks, Sensors 23 (2023) 9122, https://doi.org/10.3390/s23229122.
[25] J. Chen, C. Du, Y. Zhang, et al., A clustering-based coverage path planning method for autonomous heterogeneous UAVs, IEEE Trans. Intell. Transp. Syst. 23 (2022) 25546–25556, https://doi.org/10.1109/TITS.2021.3066240.
[26] J. Chen, F. Ling, Y. Zhang, et al., Coverage path planning of heterogeneous unmanned aerial vehicles based on ant colony system, Swarm Evolut. Comput. 69 (2022) 101005, https://doi.org/10.1016/j.swevo.2021.101005.
[27] M. Kim, J. Park, J. Park, Learning to CROSS exchange to solve min-max vehicle routing problems, Int. Conf. Learn. Represent. (2023).
[28] Sasan Mahmoudinazlou, Changhyun Kwon, A hybrid genetic algorithm for the min–max multiple traveling salesman problem, Comput. Oper. Res. 162 (2024).
[29] Y. Wang, Y. Chen, Y. Lin, Memetic algorithm based on sequential variable neighborhood descent for the minmax multiple traveling salesman problem, Comput. Ind. Eng. 106 (2017) 105–122.
[30] K. Karabulut, H. Öztop, L. Kandiller, et al., Modeling and optimization of multiple traveling salesmen problems: an evolution strategy approach, Comput. Oper. Res. 129 (2021) 105192.
[31] P. He, J. Hao, Hybrid search with neighborhood reduction for the multiple traveling salesman problem, Comput. Oper. Res. 142 (2022) 105726.
[32] J. Zheng, Y. Hong, W. Xu, et al., An effective iterated two-stage heuristic algorithm for the multiple Traveling Salesmen Problem, Comput. Oper. Res. 143 (2022) 105772.
[33] P. He, J. Hao, Memetic search for the minmax multiple traveling salesman problem with single and multiple depots, Eur. J. Oper. Res. 307 (2022) 1055–1070.
[34] M.H. Tong, Z.H. Peng, Q. Wang, A hybrid artificial bee colony algorithm with high robustness for the multiple traveling salesman problem with multiple depots, Expert Syst. Appl. 260 (2024) 125446.
[35] Pandiri Venkatesh, Alok Singh, Two metaheuristic approaches for the multiple traveling salesperson problem, Appl. Soft Comput. 26 (2015) 74–89.
[36] Banu Soylu, A general variable neighborhood search heuristic for multiple traveling salesmen problem, Comput. Ind. Eng. 90 (2015) 390–401.



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